

EXIT PROBLEMS FOR ADDITIVE-INCREASE AND MULTIPLICATIVE-DECREASE MARKOV-MODULATED PROCESSES

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1. NOTATION

Let $m \in \mathbb{N}$, given a vector $\mathbf{a} \in \mathbb{R}^m$ and a scalar $b \in \mathbb{R}$ we write $\mathbf{a} + b$ to mean $\mathbf{a} + b\mathbf{1}$, with $\mathbf{1}$ the vector with all components equal to 1. Given two vectors $\mathbf{a}, \mathbf{a}' \in \mathbb{R}^m$ we denote by $\mathbf{a} \cdot \mathbf{a}'$ the scalar vector, and by $\mathbf{a}\mathbf{a}'$ the component-wise product, that is $(\mathbf{a}\mathbf{a}')_i = a_i a'_i$. Given a function f on the real numbers, we define $f(\mathbf{a})$ as the vector with components $f(a_i)$, for example $(e^{\mathbf{a}})_i = e^{a_i}$ and $(1/\mathbf{a})_i = 1/a_i$. We set $\mathbf{a}/\mathbf{a}' = \mathbf{a}(1/\mathbf{a}')$, for $\mathbf{a}, \mathbf{a}' \in \mathbb{R}^m$. Sometimes, for aesthetic reasons, we write $a(i)$ to denote the component a_i of the vector $\mathbf{a}$.

Similarly, for a matrix $\mathbf{A} \in \mathbb{R}^{m \times m}$, we write $\mathbf{A} + b$ to mean $\mathbf{A} + b\mathbf{I}$ with $\mathbf{I}$ the identity matrix. We write $\Delta(\mathbf{a})$ or $\Delta_{\mathbf{a}}$ to denote the diagonal matrix, whose diagonal is equal to the vector $\mathbf{a}$, and we define $\mathcal{E}(\mathbf{a}) = \Delta(e^{\mathbf{a}})$. We define $|\mathbf{A}| := \|\mathbf{A}\|_{\infty} = \max_i \sum_j |\mathbf{A}_{ij}|$. With $(\cdot < b)$, we denote the indicator function of the interval $(-\infty, b)$, with $(\cdot < \mathbf{a})$ we denote the vector function with components $(\cdot < \mathbf{a})_i = (\cdot < a_i)$, and with $\Delta(\cdot < \mathbf{a})$ the matrix function with components $(\Delta(\cdot < \mathbf{a}))_{ij} = (\cdot < a_i)\delta_{ij}$. Finally we define $\mathbf{e}_i$, as the i -th row vector of the canonical base. That is $(\mathbf{e}_i)_j = \delta_{ij}$.

With $\mathbf{c}, \boldsymbol{\lambda} \in \mathbb{R}_+^m$, $\mathbf{p} \in (0, 1)^m$ and $\mathbf{Q} \in \mathbb{R}^{m \times m}$ a stochastic matrix, we define the continuous time process $\mathbf{X} := (X, J) = ((X(t), J(t)))_{t \geq 0}$, where $X(t) \in \mathbb{R}_+^m$ and $J(t) \in [1, m] \cap \mathbb{N}$, for any $t \geq 0$. The first component satisfies the following SDE

$$dX_t = \frac{1}{c(J(t))} dt + p(J(t-))X(t-)dN(t),$$

where $N(t)$ is a Markov-modulated Poisson process with rate $\lambda(J(t-))$. Denoting by T_n the n -th jump epoch of $N(t)$, that is $\Delta N(T_n) := N(T_n) - N(T_n-) = 1$ and $N(t) = n$ for $T_n \leq t < T_{n+1}$, the second component $J(t)$ is a Markov chain, defined as $J(t) = \sum_n J_n 1\{N(t) = n\}$, where $\mathbb{P}(J_{n+1} = j | J_n = i, \mathcal{F}_n^J) = q_{ij}$. We have denoted by $\mathcal{F}_n^J = \sigma(J_0, \dots, J_n)$ the sigma algebra generated by the first $(n+1)$ values attained by the modulating component $J(t)$, and with $\mathcal{F}_t = \sigma((X(s), J(s)), 0 \leq s \leq t)$, we denote by $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$, the minimal right-continuous and complete filtration generated by the Markov modulated process (X, J) .

We define the stopping times $\tau_{\uparrow a} = \inf\{t \geq 0 : X_t \geq a\}$ and $\tau_{\downarrow b} = \inf\{t \geq 0 : X_t \leq b\}$, and we set $\tau_{x \uparrow a} = (\tau_{\uparrow a} | X_0 = x)$ and $\tau_{x \uparrow a}^i = (\tau_{\uparrow a} | X_0 = x, J_0 = i)$.

We define $\mathbf{Z}(w, x \uparrow^a) := \mathbb{E}_x[e^{-w\tau_{\uparrow a}}]$ as the matrix whose ij -entry is equal to

$$(1) \quad \mathbf{Z}_{ij}(w, x \uparrow^a) := \mathbb{E}_{x,i}[e^{-w\tau_{\uparrow a}}; J(\tau_{\uparrow a}) = j].$$

To simplify notation in the following we drop the explicit reference to the parameter w of the Laplace transform, and we write

$$(2) \quad \mathbf{Z}(x \uparrow^a) := \mathbf{Z}(w, x \uparrow^a)$$

and we also define the following quantities

$$(3) \quad z_i := z_i(w) = c_i(\lambda_i + w)$$

$$(4) \quad \gamma_i := c_i \lambda_i / p_i$$

together with the vectors $\mathbf{z} := \mathbf{z}(w) = \mathbf{c}(\boldsymbol{\lambda} + w)$ and $\boldsymbol{\gamma} := \mathbf{c}\boldsymbol{\lambda}/\mathbf{p}$.

2. UPWARD ONE-SIDED PROBLEM

Theorem 2.1 (Upward one-sided problem). *Given $X_0 = x \in (0, a)$, the Laplace-Stieltjes transform of the upward exit time $\tau_{\uparrow a}$ of the additive-increase and multiplicative-decrease process $(X_t)_{t \geq 0}$ is given by*

$$(5) \quad \mathbf{Z}(x \uparrow^a) = \mathbf{Z}^{-1}(0 \uparrow^x) \mathbf{Z}(0 \uparrow^a).$$

Proof. Using the identity $\tau_{0 \uparrow a} = \tau_{0 \uparrow x} + \tau_{x \uparrow a}$ for any $x \in (0, a)$, where $\tau_{x \uparrow a}$ is conditionally independent of $\tau_{0 \uparrow x}$ given the value of $J(\tau_{0 \uparrow x})$, we have

$$\begin{aligned} \mathbf{Z}_{ij}(0 \uparrow^a) &= \mathbb{E}_{0,i}[e^{-w\tau_{\uparrow a}}; J(\tau_{\uparrow a}) = j] \\ &= \mathbb{E}_{0,i}[\mathbb{E}_{0,i}[e^{-w\tau_{\uparrow a}}; J(\tau_{\uparrow a}) = j | \tau_{\uparrow x}, J(\tau_{\uparrow x})]] \\ &= \mathbb{E}_{0,i}[e^{-w\tau_{\uparrow x}} \mathbb{E}_{0,i}[e^{-w(\tau_{\uparrow a} - \tau_{\uparrow x})}; J(\tau_{\uparrow a}) = j | \tau_{\uparrow x}, J(\tau_{\uparrow x})]] \\ &= \mathbb{E}_{0,i}[e^{-w\tau_{\uparrow x}} \mathbb{E}_{x, J(\tau_{\uparrow x})}[e^{-w\tau_{\uparrow a}}; J(\tau_{\uparrow a}) = j]] \\ &= \sum_k \mathbb{E}_{0,i}[e^{-w\tau_{\uparrow x}}; J(\tau_{\uparrow x}) = k] \mathbb{E}_{x,k}[e^{-w\tau_{\uparrow a}}; J(\tau_{\uparrow a}) = j] \\ &= \sum_k \mathbf{Z}_{ik}(0 \uparrow^x) \mathbf{Z}_{kj}(x \uparrow^a), \end{aligned}$$

or in matrix notation

$$(6) \quad \mathbf{Z}(0 \uparrow^a) = \mathbf{Z}(0 \uparrow^x) \mathbf{Z}(x \uparrow^a).$$

To show that $\mathbf{Z}(0 \uparrow^x)$ is invertible, we use a standard local invertibility argument for continuous evolution families of matrices, see Engel and Nagel [2000]. By continuity and the condition $\mathbf{Z}(x \uparrow^x) = I$, for every $x \geq 0$ there exists $\varepsilon_x > 0$ such that $\mathbf{Z}(x \uparrow^a)$ is invertible whenever $a - x < \varepsilon_x$. Since $[0, x]$ is compact, we may choose a partition $0 = x_0 < x_1 < \dots < x_N = x$ such that each $\mathbf{Z}(x_i \uparrow^{x_{i+1}})$ is invertible. The identity (6) then yields

$$\mathbf{Z}(0 \uparrow^x) = \mathbf{Z}(x_0 \uparrow^{x_1}) \mathbf{Z}(x_1 \uparrow^{x_2}) \cdots \mathbf{Z}(x_{N-1} \uparrow^{x_N}),$$

and therefore $\mathbf{Z}(0 \uparrow^x)$ is invertible, and the result holds. $\square$

With $T = \inf\{t \geq 0, \Delta X_t < 0\}$ and, on $\{T \leq a c(J_0)\}$, let $Y = Tp(J_0)/c(J_0)$, we may split the hitting time $\tau_{0 \uparrow a}$ in the following way

$$(7) \quad \tau_{0 \uparrow a} = a c(J_0)(T > a c(J_0)) + (T + \tau_{Y \uparrow a})(T \leq a c(J_0))$$

It follows that

$$\begin{aligned} \mathbf{Z}_{ij}(0 \uparrow^a) &= e^{-c_i(\lambda_i + w)a} \delta_{ij} + \int_0^{ac_i} \lambda_i e^{-(\lambda_i + w)t} \sum_k q_{ik} \mathbb{E}_{t p_i / c_i, k}[e^{-w\tau_{\uparrow a}}; J(\tau_{\uparrow a}) = j] dt \\ &= e^{-az_i} \delta_{ij} + \int_0^\infty (x \leq p_i a) \gamma_i e^{-\frac{z_i}{p_i} x} \sum_k q_{ik} \mathbf{Z}_{kj}(x \uparrow^a) dx, \end{aligned}$$

in matrix form

$$\begin{aligned} \mathbf{Z}(0 \uparrow^a) &= \mathcal{E}(-az) + \Delta(\gamma) \int_0^\infty \Delta(x \leq pa) \mathcal{E}(-(z/p)x) \mathbf{Q} \mathbf{Z}(x \uparrow^a) dx \\ &= \mathcal{E}(-az) + \Delta(\gamma) \int_0^\infty \Delta(x \leq pa) \mathcal{E}(-(z/p)x) \mathbf{Q} \mathbf{Z}^{-1}(0 \uparrow^x) \mathbf{Z}(0 \uparrow^a) dx, \end{aligned}$$

and multiplying on the right both sides by $\mathbf{U}(a) := \mathbf{Z}^{-1}(0 \uparrow^a)$ we get

$$\mathbf{I} = \mathcal{E}(-az) \mathbf{U}(a) + \Delta(\gamma) \int_0^\infty \Delta(x \leq pa) \mathcal{E}(-(z/p)x) \mathbf{Q} \mathbf{U}(x) dx.$$

Multiply the equation by $e^{-za} \mathcal{E}(az)$, we get

$$\begin{aligned} \mathcal{E}(a(z-z)) &= e^{-za} \mathbf{U}(a) \\ &+ \mathcal{E}(a(z-z)) \Delta(\gamma) \int_0^\infty \Delta(x \leq pa) \mathcal{E}(-(z/p)x) \mathbf{Q} \mathbf{U}(x) dx. \end{aligned}$$

Then integrating on $a \in [0, \infty]$, we have

$$\begin{aligned} \int_0^\infty \mathcal{E}(a(z-z)) da &= \int_0^\infty e^{-za} \mathbf{U}(a) da \\ &+ \Delta(\gamma) \int_0^\infty \left(\int_{x/p}^\infty \mathcal{E}(a(z-z)) da \right) \mathcal{E}(-(z/p)x) \mathbf{Q} \mathbf{U}(x) dx. \end{aligned}$$

Defining $\tilde{\mathbf{U}}(z) := \int_0^\infty e^{-za} \mathbf{U}(a) da$, since $\int_x^\infty \mathcal{E}(-az) dz = \Delta^{-1}(\mathbf{a}) \mathcal{E}(-\mathbf{a}x)$,

$$\mathbf{I} = \Delta(z-z) \tilde{\mathbf{U}}(z) + \Delta(\gamma) \int_0^\infty \mathcal{E}(-zx/p) \mathbf{Q} \mathbf{U}(x) dx.$$

By writing $\mathcal{E}(-(z/p)x) = \sum_i \mathbf{E}_i e^{-zx/p_i}$, with $\mathbf{E}_k = (\delta_{ik} \delta_{kj})_{ij}$, we get

$$\begin{aligned} \mathbf{I} &= \Delta(z-z) \tilde{\mathbf{U}}(z) + \Delta(\gamma) \sum_i \mathbf{E}_i \mathbf{Q} \int_0^\infty e^{-zx/p_i} \mathbf{U}(x) dx \\ &= \Delta(z-z) \tilde{\mathbf{U}}(z) + \Delta(\gamma) \sum_i \mathbf{E}_i \mathbf{Q} \tilde{\mathbf{U}}(z/p_i) \end{aligned}$$

With $\mathbf{A}(z) := \Delta(\gamma/(z-z))$ and $\mathbf{B} := -\Delta^{-1}(\gamma)$, we have

$$(8) \quad \tilde{\mathbf{U}}(z) = \mathbf{A}(z) \mathbf{B} + \mathbf{A}(z) \sum_i \mathbf{E}_i \mathbf{Q} \tilde{\mathbf{U}}(z/p_i).$$

This equation may be solved by iteration, indeed we have

$$\begin{aligned} \tilde{\mathbf{U}}(z) &= \mathbf{A}(z) \mathbf{B} + \sum_{i_1} \mathbf{A}(z) \mathbf{E}_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) \mathbf{B} \\ &+ \mathbf{A}(z) \sum_{i_1, i_2} \mathbf{E}_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) \mathbf{E}_{i_2} \mathbf{Q} \tilde{\mathbf{U}}(z/(p_{i_1} p_{i_2})) \\ &= \mathbf{A}(z) \mathbf{B} + \sum_{i_1} \mathbf{A}(z) \mathbf{E}_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) \mathbf{B} \\ &+ \sum_{i_1, i_2} \mathbf{A}(z) \mathbf{E}_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) \mathbf{E}_{i_2} \mathbf{Q} \mathbf{A}(z/(p_{i_1} p_{i_2})) \mathbf{B} \\ &+ \sum_{i_1, i_2} \mathbf{A}(z) \mathbf{E}_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) \mathbf{E}_{i_2} \mathbf{Q} \tilde{\mathbf{U}}(z/(p_{i_1} p_{i_2})) \end{aligned}$$

With $\mathbf{D}(z; i_1, i_2, \dots, i_n) = \mathbf{E}_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) \cdots \mathbf{E}_{i_n} \mathbf{Q} \mathbf{A}(z/(p_{i_1} \cdots p_{i_n}))$, we define

$$(9) \quad D(z) = \mathbf{I} + \sum_{n \geq 1} \sum_{i_1, \dots, i_n} \mathbf{D}(z; i_n, \dots, i_1).$$

Remark 2.2. Notice that the definition in (9) is well posed as the series is convergent. Indeed $|\mathbf{A}(z/(p_{i_1} \cdots p_{i_n}))| \leq p_{\max}^n (\gamma_{\max}/z)$, so that

$$|\mathbf{D}(z; i_1, i_2, \dots, i_n)| \leq (p_{\max}^{(n+1)/2} \gamma_{\max} z^{-1})^n$$

and, by triangular inequality,

$$\left| \sum_{i_1, \dots, i_n} \mathbf{D}(z; i_n, \dots, i_1) \right| \leq (p_{\max}^{(n+1)/2} m \gamma_{\max} z^{-1})^n \leq (1/2)^n,$$

where the last equality is valid for any $n > n_0(z)$ for some $n_0(z) \in \mathbb{N}$. This implies that the series that defines $D(z)$ is absolutely convergent for any $z \in \mathbb{C}$.

We may rewrite the expression before as

$$(10) \quad \tilde{\mathbf{U}}(z) = \mathbf{A}(z) D(z) \mathbf{B}.$$

2.1. Solution of (8). An alternative expression for the solution of (8) is given by

$$\begin{aligned} \gamma_j \tilde{\mathbf{U}}_{ij}(z) &= a_i(z) \delta_{ij} + a_i(z) a_j(z/p_i) q_{ij} \\ &\quad + a_i(z) \sum_{n=1}^{\infty} \sum_{\substack{\kappa_1, \dots, \kappa_n \\ k_0=i, k_{n+1}=j}} \prod_{m=1}^{n+1} (a_{\kappa_m}(z/(p_i p_{\kappa_1} \cdots p_{\kappa_{m-1}})) q_{\kappa_{m-1}, \kappa_m}) \end{aligned}$$

where $a_i(z) := a_i(w, z) = \gamma_i/(z_i - z)$.

3. DOWNWARD ONE-SIDED PROBLEM

With $T = \inf\{t \geq 0, \Delta X_t < 0\}$, $c = c(J_0)$ and $p = p(J_0)$,

$$(11) \quad \tau_{x \downarrow b} = (T + \tau_{(x+T/c)p \downarrow b}) \mathbb{1}\{x > b\}.$$

We define the Laplace of $\tau_{x \downarrow b}$ as

$$(12) \quad \mathbf{Z}_{ij}(x \downarrow b) := \mathbf{Z}_{ij}(w, x \downarrow b) = \mathbb{E}_{x,i}[e^{-w\tau_{x \downarrow b}}; J(\tau_{x \downarrow b}) = j].$$

We have that, for $x \leq b$, $\mathbf{Z}(x \downarrow b) = \mathbf{I}$. Let $x > b$, then

$$\begin{aligned} \mathbf{Z}_{ij}(x \downarrow b) &= \mathbb{E}_{x,i}[e^{-wT} e^{-w\tau_{(x+T/c)p_i \downarrow b}}; J(\tau_{x \downarrow b}) = j] \\ &= \mathbb{E}_{x,i}[e^{-wT} \mathbb{E}_{x,i}[e^{-w\tau_{(x+T/c)p_i \downarrow b}}; J(\tau_{x \downarrow b}) = j|T]] \\ &= \mathbb{E}_{x,i}[e^{-wT} \mathbb{E}_{(x+T/c)p_i, J(T)}[e^{-w\tau_{x \downarrow b}}; J(\tau_{x \downarrow b}) = j]] \\ &= \mathbb{E}_{x,i}[e^{-wT} \mathbf{Z}_{J(T)j}((x+T/c)p_i \downarrow b)]. \end{aligned}$$

Since on $\{J(0) = i\}$, $T \sim \text{Exp}(\lambda_i)$ and $\mathbb{P}(J(T) = k) = q_{ik}$, we have

$$\begin{aligned} \mathbf{Z}_{ij}(x \downarrow_b) &= \sum_k q_{ik} \mathbb{E}_{x,i} [e^{-wT} \mathbf{Z}_{kj}((x + T/c_i)p_i \downarrow_b)] \\ &= \sum_k q_{ik} \int_0^\infty \mathbf{Z}_{kj}((x + t/c_i)p_i \downarrow_b) \lambda_i e^{-(w+\lambda_i)t} dt \\ &= \sum_k q_{ik} \gamma_i e^{z_i x} \int_{p_i x}^\infty \mathbf{Z}_{kj}(y \downarrow_b) e^{-(z_i/p_i)y} dy \end{aligned}$$

Let $\tilde{\mathbf{Z}}(z) = \int_0^\infty e^{-zx} \mathbf{Z}(x \downarrow_b) dx$, then

$$\begin{aligned} \tilde{\mathbf{Z}}_{ij}(z) &= \gamma_i \sum_k q_{ik} \int_0^\infty e^{(z_i-z)x} \int_{p_i x}^\infty \mathbf{Z}_{kj}(y \downarrow_b) e^{-(z_i/p_i)y} dy dx \\ &= \gamma_i \sum_k q_{ik} \int_0^\infty \mathbf{Z}_{kj}(y \downarrow_b) e^{-(z_i/p_i)y} \left(\int_0^{y/p_i} e^{(z_i-z)x} dx \right) dy \\ &= \frac{\gamma_i}{z_i - z} \sum_k q_{ik} \int_0^\infty \mathbf{Z}_{kj}(y \downarrow_b) (e^{-(z/p_i)y} - e^{-(z_i/p_i)y}) dy \\ &= \sum_k \frac{\gamma_i q_{ik}}{z_i - z} \left(\tilde{\mathbf{Z}}_{kj}(z/p_i) - \tilde{\mathbf{Z}}_{kj}(z_i/p_i) \right). \end{aligned}$$

As before, with $\gamma_i = c_i \lambda_i / p_i$ and $z_i = c_i(\lambda_i + w)$ as in (3-4), we let again $a_i(z) = \gamma_i / (z_i - z)$,

$$\begin{aligned} \tilde{\mathbf{Z}}_{ij}(z) &= \sum_k a_i(z) q_{ik} \left(\tilde{\mathbf{Z}}_{kj}(z/p_i) - \tilde{\mathbf{Z}}_{kj}(z_i/p_i) \right) \\ &= \sum_h \sum_l a_i(z) \delta_{ih} \delta_{hl} \sum_k q_{lk} \left(\tilde{\mathbf{Z}}_{kj}(z/p_h) - \tilde{\mathbf{Z}}_{kj}(z_h/p_h) \right) \end{aligned}$$

so that, with the above defined $\mathbf{A}(z) = \mathbf{\Delta}^{-1}(\mathbf{z} - z)\mathbf{\Delta}(\boldsymbol{\gamma}) = \mathbf{\Delta}(\mathbf{a}(z))$ and $\mathbf{E}_k = (\delta_{ik}\delta_{kj})_{i,j}$, we can write the previous expression in matrix form

$$\tilde{\mathbf{Z}}(z) = \sum_h \mathbf{A}(z) \mathbf{E}_h \mathbf{Q} \left(\tilde{\mathbf{Z}}(z/p_h) - \tilde{\mathbf{Z}}(z_h/p_h) \right).$$

Then, recursively, and relabelling in decreasing order the indexes of the sums, that is $\dots, i_n, i_{n-1}, \dots, i_2, i_1$, we get

$$\begin{aligned}
\tilde{\mathbf{Z}}(z) &= \sum_{i_1} \mathbf{A}(z) E_{i_1} \mathbf{Q} \tilde{\mathbf{Z}}(z/p_{i_1}) - \sum_i \mathbf{A}(z) E_i \mathbf{Q} \tilde{\mathbf{Z}}(z_i/p_i) \\
&= \sum_{i_1, i_2} \mathbf{A}(z) E_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) E_{i_2} \mathbf{Q} \tilde{\mathbf{Z}}(z/(p_{i_1} p_{i_2})) \\
&\quad - \mathbf{A}(z) \sum_i \left(\sum_{i_1} E_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) + \mathbf{I} \right) E_i \mathbf{Q} \tilde{\mathbf{Z}}(z_i/p_i) \\
&= \sum_{i_1, i_2, i_3} \mathbf{A}(z) E_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) E_{i_2} \mathbf{Q} \mathbf{A}(z/(p_{i_1} p_{i_2})) E_{i_3} \mathbf{Q} \tilde{\mathbf{Z}}(z/(p_{i_1} p_{i_2} p_{i_3})) \\
&\quad - \mathbf{A}(z) \sum_i \left(\sum_{i_1, i_2} E_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) E_{i_2} \mathbf{Q} \mathbf{A}(z/(p_{i_1} p_{i_2})) + \sum_{i_1} E_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) + \mathbf{I} \right) E_i \mathbf{Q} \tilde{\mathbf{Z}}(z_i/p_i) \\
&= \sum_{i_1, i_2, i_3} \mathbf{A}(z) E_{i_1} \mathbf{Q} \mathbf{A}(z/p_{i_1}) E_{i_2} \mathbf{Q} \mathbf{A}(z/(p_{i_1} p_{i_2})) E_{i_3} \mathbf{Q} \tilde{\mathbf{Z}}(z/(p_{i_1} p_{i_2} p_{i_3})) \\
&\quad - \mathbf{A}(z) \sum_i \left(\sum_{i_1, i_2} D(z; i_2, i_1) + \sum_{i_1} D(z; i_1) + \mathbf{I} \right) E_i \mathbf{Q} \tilde{\mathbf{Z}}(z_i/p_i) \\
&= -\mathbf{A}(z) \sum_i D(z) E_i \mathbf{Q} \tilde{\mathbf{Z}}(z_i/p_i)
\end{aligned}$$

where $D(z)$ has been defined in (9).

Therefore, with $\mathbf{H} = -\sum_i E_i \mathbf{Q} \tilde{\mathbf{Z}}(z_i/p_i)$, we may write

$$(13) \quad \tilde{\mathbf{Z}}(z) = \mathbf{A}(z) D(z) \mathbf{H}$$

Note: It is left to show that the limit of the reminder goes to 0.

Multiplying the expression in (13) on the right by $\Delta^{-1}(\mathbf{z} - z)$ we obtain

$$\Delta^{-1}(\mathbf{z} - z) \tilde{\mathbf{Z}}(z) = \Delta(\gamma) D(z) \mathbf{H}$$

Note: It is left to show how to compute the entries of the matrix $\mathbf{H}$. Probably taking the limit for $z \rightarrow z_i/p_i$

4. TWO SIDED EXIT PROBLEMS

With $0 \leq b < a$, we define $\mathbf{L}(w, x \uparrow_b^a) := \mathbb{E}_x[e^{-w\tau_{\uparrow a}}; \tau_{\uparrow a} < \tau_{\downarrow b}]$ as the matrix whose ij -entry is equal to

$$(14) \quad \mathbf{L}_{ij}(w, x \uparrow_b^a) := \mathbb{E}_{x,i}[e^{-w\tau_{\uparrow a}}; \tau_{\uparrow a} < \tau_{\downarrow b}, J(\tau_{\uparrow a}) = j],$$

for brevity we set

$$(15) \quad \mathbf{L}(x \uparrow_b^a) := \mathbf{L}(w, x \uparrow_b^a).$$

Let $\tau_{a,b} = \tau_{\uparrow a} \wedge \tau_{\downarrow b}$, since we have $X(\tau_{a,b}) \in \{a, b\}$, we get

$$(16) \quad \mathbf{L}(X(\tau_{a,b}) \uparrow_b^a) = (\tau_{\uparrow a} < \tau_{\downarrow b}) \mathbf{I}.$$

Proposition 4.1. *We have that, for any $x \in [b, b/p)$, where $1/p = \min_i 1/p_i$,*

$$(17) \quad \mathbf{L}(b \uparrow_b^x) = \mathcal{E}(-(x - b)\mathbf{z}),$$

$$(18) \quad \mathbf{L}(x \uparrow_b^{b/p}) = \mathcal{E}(-(b/p - x)\mathbf{z}).$$

Proof. Assume that $(x_0, J_0) = (b, i)$, then, since $x p_i < b$, we have that $\{\tau_{\uparrow x} < \tau_{\downarrow b}\} = \{c_i T > (x - b)\}$. Therefore

$$(19) \quad \mathbf{L}_{ij}(b \uparrow_b^x) = \delta_{ij} \mathbb{P}_{x,i}(\tau_{\uparrow x} < \tau_{\downarrow b}) = \delta_{ij} \mathbb{P}_{x,i}(c_i T \geq (x - b)) = \delta_{ij} \exp\{-(z_i/p)(x - b)\}.$$

Similarly, if $(x_0, J_0) = (x, i)$, then $\{\tau_{\uparrow b/p} < \tau_{\downarrow b}\} = \{c_i T \geq (b/p - x)\}$, so the result follows. $\square$

Proposition 4.2. *Let $b \leq x \leq a' < a$, then*

$$(20) \quad \mathbf{L}(x \uparrow_b^a) = \mathbf{L}(x \uparrow_b^{a'}) \mathbf{L}(a' \uparrow_b^a).$$

In addition, for any $b \leq x \leq a$, $\mathbf{L}(x \uparrow_b^a)$ is invertible.

Proof. Since $\tau_{\uparrow a} \geq \tau_{\uparrow a'}$ on $\{\tau_{\uparrow a} \leq \tau_{\downarrow b}\}$ by applying the strong Markov property at time $\tau_{\uparrow a'}$ on the set $\{\tau_{\uparrow a} \leq \tau_{\downarrow b}\}$, and using the fact that the process starts anew in $(a', J(\tau_{\uparrow a'}))$, we get the result. For the invertibility, we use the standard local invertibility argument for continuous evolution families of matrices as in the proof of Theorem 2.1, see also Engel and Nagel [2000]. $\square$

Theorem 4.3. *Assume that $\mathbf{L}(x \uparrow_b^a)$ is known for all $x < a$, for some $a > 0$, then there exists $p < 1$, not depending on a , such that $\mathbf{L}(x \uparrow_b^{a/p})$ can be computed for all $x < a/p$.*

In particular,

$$(21) \quad \mathbf{L}(a \uparrow_b^{a/p}) = (\mathbf{I} - \mathbf{H}(a))^{-1} \mathcal{E}(-(a/p - a)\mathbf{z}),$$

where the matrix $\mathbf{H}(x)$ is defined in (23), for $x \in [a, a/p)$.

Then, for $x \in (a, a/p)$

$$(22) \quad \mathbf{L}(x \uparrow_b^{a/p}) = \mathcal{E}(-(a/p - x)\mathbf{z}) + \mathbf{H}(x) \mathbf{L}(a \uparrow_b^{a/p}).$$

Proof. We choose $0 < p < \min\{p_i : i \in E\}$, such that $p_i x < a$ for all $x \in [a, a/p)$.

Let $x \in [a, a/p)$, $t_i(x) = c_i(a/p - x)$, and $T \sim \text{Exp}(\lambda_i)$, then

$$\begin{aligned} \mathbf{L}_{ij}(x \uparrow_b^{a/p}) &= e^{-wt_i(x)} \delta_{ij} \mathbb{P}_{x,i}(T > t_i(x)) + \mathbb{E}_{x,i}[e^{-w\tau_{\uparrow a/p}}; J(\tau_{\uparrow a/p}) = j, \tau_{\uparrow a} < \tau_{\downarrow b}, T^i \leq t_i(x)] \\ &= e^{-(w+\lambda_i)t_i(x)} \delta_{ij} + \sum_h q_{ih} \int_0^{t_i(x)} \mathbf{L}_{hj}((x + t/c_i)p_i \uparrow_b^{a/p}) \lambda_i e^{-(w+\lambda_i)t} dt \\ &= e^{-z_i(a/p-x)} \delta_{ij} + \sum_h q_{ih} \gamma_i e^{\frac{z_i}{p_i}x} \int_{p_i x}^{p_i a/p} \mathbf{L}_{hj}(y \uparrow_b^{a/p}) e^{-\frac{z_i}{p_i}y} dy \\ &= e^{-z_i(a/p-x)} \delta_{ij} + \sum_{h'} \gamma_i e^{\frac{z_i}{p_i}x} \int_{p_i x}^{p_i a/p} \sum_h q_{ih} \mathbf{L}_{hh'}(y \uparrow_b^a) e^{-\frac{z_i}{p_i}y} dy \mathbf{L}_{h'j}(a \uparrow_b^{a/p}) \\ &= e^{-z_i(a/p-x)} \delta_{ij} + \sum_{h'} \mathbf{H}_{ih'}(x) \mathbf{L}_{h'j}(a \uparrow_b^{a/p}) \end{aligned}$$

where, for any $i, j \in E$ and $x \in [a, a/p)$, the quantity

$$(23) \quad \mathbf{H}_{ij}(x) := \gamma_i e^{\frac{z_i}{p_i}x} \int_{p_i x}^{p_i a/p} \sum_h q_{ih} \mathbf{L}_{hj}(y \uparrow_b^a) e^{-\frac{z_i}{p_i}y} dy,$$

can be computed since $\mathbf{L}_{hj}(y \uparrow_b^a)$ is known for all $y < a$ and $p_i x < p_i a/p \leq a$.

We can write the previous equation in matrix form as

$$\mathbf{L}(x \uparrow_b^{a/p}) = \mathcal{E}(-(a/p - x)\mathbf{z}) + \mathbf{H}(x) \mathbf{L}(a \uparrow_b^{a/p}).$$

and in particular, for $x = a$,

$$\mathbf{L}(a \uparrow_b^{a/p}) = (\mathbf{I} - \mathbf{H}(a))^{-1} \boldsymbol{\mathcal{E}}(-(a/p - a)\mathbf{z}).$$

□

Theorem 4.4. *The Laplace–Stieltjes transform of the downward two-sided exit time, $\mathbf{L}(x \downarrow_b^a)$, whose components for $x \in [b, a)$, are given by*

$$(24) \quad \mathbf{L}_{ij}(w, x \downarrow_b^a) := \mathbb{E}_{x,i}[e^{-w\tau_{\downarrow b}}; \tau_{\downarrow b} < \tau_{\uparrow a}, J(\tau_{\downarrow b}) = j],$$

is given by

$$(25) \quad \mathbf{L}(x \downarrow_b^a) = \mathbf{Z}(x \downarrow_b) - \mathbf{L}(x \uparrow_b^a) \mathbf{Z}(a \downarrow_b),$$

Proof. The result follows by the equality

$$\begin{aligned} \tau_{\downarrow b} &= \tau_{\downarrow b}(\tau_{\downarrow b} < \tau_{\uparrow a}) + \tau_{\downarrow b}(\tau_{\uparrow a} \leq \tau_{\downarrow b}) \\ &= \tau_{\downarrow b}(\tau_{\downarrow b} < \tau_{\uparrow a}) + (\tau_{\uparrow a} + (\tau_{\downarrow b} - \tau_{\uparrow a}))(\tau_{\uparrow a} \leq \tau_{\downarrow b}) \end{aligned}$$

and an application of the strong Markov property at time $\tau_{\uparrow a}$, on the set $\{\tau_{\uparrow a} \leq \tau_{\downarrow b}\}$, where the process assumes the value $(a, J(\tau_{\uparrow a}))$. □

5. EXIT IDENTITIES FOR REFLECTED PROCESSES

Consider the row vector $(\tau_{\uparrow a} < \tau_{\downarrow b}; J(\tau_{a,b}))$ whose components are given by

$$(26) \quad \begin{aligned} (\tau_{\uparrow a} < \tau_{\downarrow b}; J(\tau_{a,b}))_j &= (\tau_{\uparrow a} < \tau_{\downarrow b}; J(\tau_{a,b}) = j) \\ &= (\tau_{\uparrow a} < \tau_{\downarrow b}) \delta_{J(\tau_{a,b})j}. \end{aligned}$$

We have that

$$\begin{aligned} (27) \quad t \mapsto \mathbb{E}_{x,i}[e^{-w\tau_{a,b}}; \tau_{\uparrow a} < \tau_{\downarrow b}, J(\tau_{a,b}) | \mathcal{F}_t] &= \mathbb{E}_{x,i}[e^{-w\tau_{a,b}}; t < \tau_{a,b}, \tau_{\uparrow a} < \tau_{\downarrow b}, J(\tau_{a,b}) | \mathcal{F}_t] \\ &\quad + \mathbb{E}_{x,i}[e^{-w\tau_{a,b}}; \tau_{a,b} \leq t, \tau_{\uparrow a} < \tau_{\downarrow b}, J(\tau_{a,b}) | \mathcal{F}_t] \\ &= (t < \tau_{a,b}) e^{-wt} \mathbb{E}_{X(t), J(t)}[e^{-w\hat{\tau}_{a,b}}; \hat{\tau}_{\uparrow a} < \hat{\tau}_{\downarrow b}, J(\hat{\tau}_{a,b})] \\ &\quad + (\tau_{a,b} \leq t) e^{-w\tau_{a,b}} (\tau_{\uparrow a} < \tau_{\downarrow b}; J(\tau_{a,b})) \\ &= (t < \tau_{a,b}) e^{-wt} \mathbf{e}_{J(\tau_{a,b})} \mathbf{L}(X(t) \uparrow_b^a) \\ &\quad + (\tau_{a,b} \leq t) e^{-w\tau_{a,b}} \mathbf{e}_{J(\tau_{a,b})} \mathbf{L}(X(\tau_{a,b}) \uparrow_b^a) \\ &= e^{-w(\tau_{a,b} \wedge t)} \mathbf{e}_{J(\tau_{a,b} \wedge t)} \mathbf{L}(X(\tau_{a,b} \wedge t) \uparrow_b^a) \end{aligned}$$

is a local martingale. In the second inequality, we used the Markov property of the Markov-modulated process (X, J) , and in the third equality we applied the definition (14).

In the same way, from (12), consider the random row vector $\mathbf{e}_{J(\tau_{a,b})} \mathbf{Z}(X(\tau_{a,b}) \downarrow_b)$

$$(28) \quad (\mathbf{e}_{J(\tau_{a,b})} \mathbf{Z}(X(\tau_{a,b}) \downarrow_b))_j = \mathbb{E}_{X(\tau_{a,b}), J(\tau_{a,b})}[e^{-w\tau_{\downarrow b}}; J(\tau_{\downarrow b}) = j].$$

We have that

$$\begin{aligned}
(29) \quad t \mapsto \mathbb{E}_{x,i}[e^{-w\tau_{a,b}} \mathbf{e}_{J(\tau_{a,b})} \mathbf{Z}(X(\tau_{a,b}) \downarrow_b) | \mathcal{F}_t] &= \mathbb{E}_{x,i}[e^{-w\tau_{a,b}} \mathbf{e}_{J(\tau_{a,b})} \mathbf{Z}(X(\tau_{a,b}) \downarrow_b); t < \tau_{a,b} | \mathcal{F}_t] \\
&\quad + (\tau_{a,b} \leq t) e^{-w\tau_{a,b}} \mathbf{e}_{J(\tau_{a,b})} \mathbf{Z}(X(\tau_{a,b}) \downarrow_b) \\
&= \mathbb{E}_{x,i}[e^{-w\tau_{a,b}} \mathbb{E}_{X(\tau_{a,b}), J(\tau_{a,b})}[e^{-w\hat{\tau}_{\downarrow b}}; J(\hat{\tau}_{\downarrow b})]; t < \tau_{a,b} | \mathcal{F}_t] \\
&\quad + (\tau_{a,b} \leq t) e^{-w\tau_{a,b}} \mathbf{e}_{J(\tau_{a,b})} \mathbf{Z}(X(\tau_{a,b}) \downarrow_b) \\
&= \mathbb{E}_{x,i}[\mathbb{E}_{X(\tau_{a,b}), J(\tau_{a,b})}[e^{-w(\tau_{a,b} + \hat{\tau}_{\downarrow b})}; J(\hat{\tau}_{\downarrow b})]; t < \tau_{a,b} | \mathcal{F}_t] \\
&\quad + (\tau_{a,b} \leq t) e^{-w\tau_{a,b}} \mathbf{e}_{J(\tau_{a,b})} \mathbf{Z}(X(\tau_{a,b}) \downarrow_b) \\
&= \mathbb{E}_{x,i}[e^{-w\tau_{\downarrow b}}; J(\tau_{\downarrow b}); t < \tau_{a,b} | \mathcal{F}_t] \\
&\quad + (\tau_{a,b} \leq t) e^{-w\tau_{a,b}} \mathbf{e}_{J(\tau_{a,b})} \mathbf{Z}(X(\tau_{a,b}) \downarrow_b) \\
&= (t < \tau_{a,b}) e^{-wt} \mathbb{E}_{X(t), J(t)}[e^{-w\tilde{\tau}_{\downarrow b}}; J(\tilde{\tau}_{\downarrow b})] \\
&\quad + (\tau_{a,b} \leq t) e^{-w\tau_{a,b}} \mathbf{e}_{J(\tau_{a,b})} \mathbf{Z}(X(\tau_{a,b}) \downarrow_b) \\
&= (t < \tau_{a,b}) e^{-wt} \mathbf{e}_{J(t)} \mathbf{Z}(X(t) \downarrow_b) \\
&\quad + (\tau_{a,b} \leq t) e^{-w\tau_{a,b}} \mathbf{e}_{J(\tau_{a,b})} \mathbf{Z}(X(\tau_{a,b}) \downarrow_b) \\
&= e^{-w(\tau_{a,b} \wedge t)} \mathbf{e}_{J(\tau_{a,b} \wedge t)} \mathbf{Z}(X(\tau_{a,b} \wedge t) \downarrow_b)
\end{aligned}$$

is a local martingale. Looking at (29), in the fifth equality we used the Markov property of the Markov-modulated process (X, J) and in the sixth one we used the definition (12).

Let

$$(30) \quad \mathbf{F}(w, x \uparrow_b^a) := \mathbf{Z}(w, x \downarrow_b) + \mathbf{L}(w, x \uparrow_b^a),$$

by linear combination, we have that the row vector function

$$(31) \quad t \mapsto e^{-w(\tau_{a,b} \wedge t)} \mathbf{e}_{J(\tau_{a,b} \wedge t)} \mathbf{F}(w, X(\tau_{a,b} \wedge t) \uparrow_b^a)$$

is a local martingale.

Define $\underline{X}(t) = \underline{X}(0) \wedge \inf_{s \leq t} X(s)$ and let $\hat{X}(t) = X(t) - \underline{X}(t)$ and consider the Markov-modulated process $(\underline{X}(t), \hat{X}(t), J(t))$, corresponding to the initial process $(X(t), J(t))$, reflected at its infimum. For this process, we want to study the exit time from a barrier $c > 0$, that is the distribution of the stopping time $\hat{\tau}_c = \inf\{t \geq 0, \hat{X}(t) \geq c\}$. And in particular we want to compute, for $v \leq x \leq v + c$, with $\underline{X}(0) = v > 0$, the laplace transform of $\hat{\tau}_c$ defined as the matrix $\hat{\mathbf{Z}}(w, v, x \uparrow^c)$ whose ij -entry is given by

$$(32) \quad \hat{\mathbf{Z}}_{ij}(w, v, x \uparrow^c) = \mathbb{E}_{v,x,i}[e^{-w\hat{\tau}_c}; J(\hat{\tau}_c) = j].$$

We are going to show that

$$(33) \quad \mathbb{E}_{v,x,i}[e^{-w\hat{\tau}_c}; J(\hat{\tau}_c) = j] = \mathbf{F}_{ij}(w, x \uparrow_{\mathbf{B}_{ij}(w,v,c)}^{\mathbf{A}_{ij}(w,v,c)}),$$

or, in more concise matrix notation, equivalently

$$(34) \quad \mathbb{E}_{v,x}[e^{-w\hat{\tau}_c}; J(\hat{\tau}_c)] = \mathbf{F}(w, x \uparrow_{\mathbf{B}(w,v,c)}^{\mathbf{A}(w,v,c)}).$$

Let us first focus on the left side of (33) we have that

$$(35) \quad \mathbb{E}_{v,x,i}[e^{-w\hat{\tau}_c}; J(\hat{\tau}_c) = j] = \mathbb{E}_{v,x,i}[e^{-w\hat{\tau}_c} \delta_{J(\hat{\tau}_c)j}],$$

so in the same notation of (34), since $(\mathbf{e}_{J(\hat{\tau}_c)})_j = \delta_{J(\hat{\tau}_c)j}$, we can write

$$(36) \quad \mathbb{E}_{v,x,i}[e^{-w\hat{\tau}_c}; J(\hat{\tau}_c)] = \mathbb{E}_{v,x,i}[e^{-w\hat{\tau}_c} \mathbf{e}_{J(\hat{\tau}_c)}],$$

therefore we may rewrite (34) as

$$(37) \quad \mathbb{E}_{v,x,i}[e^{-w\hat{\tau}_c} \mathbf{e}_{J(\hat{\tau}_c)}] = \mathbf{e}_i \mathbf{F}(w, x \uparrow_{\mathbf{B}(w,v,c)}^{\mathbf{A}(w,v,c)}).$$

This means that we would like that the row vector function

$$(38) \quad t \rightarrow e^{-wt} \mathbf{e}_{J(t)} \mathbf{F}(w, X(t) \uparrow_{\underline{X}(t)}^{\mathbf{A}(w,\underline{X}(t),c)})$$

is a martingale, such that, by the optional stopping theorem, we get that

$$(39) \quad \mathbb{E}_{v,x,i}[e^{-w\hat{\tau}_c} \mathbf{e}_{J(\hat{\tau}_c)} \mathbf{F}(w, X(\hat{\tau}_c) \uparrow_{\mathbf{B}(w,\underline{X}(\hat{\tau}_c),c)}^{\mathbf{A}(w,\underline{X}(\hat{\tau}_c),c)})] = \mathbf{e}_i \mathbf{F}(w, x \uparrow_{\mathbf{B}(w,v,c)}^{\mathbf{A}(w,v,c)}).$$

In particular we would like that, a.s., on the event $\{\hat{\tau}_c < \infty\}$, we have that

$$(40) \quad \mathbf{F}(w, X(\hat{\tau}_c) \uparrow_{\mathbf{B}(w,\underline{X}(\hat{\tau}_c),c)}^{\mathbf{A}(w,\underline{X}(\hat{\tau}_c),c)}) = \mathbf{I},$$

such that (39) reduces to (37), and get that

$$(41) \quad \hat{\mathbf{Z}}(w, v, x \uparrow^c) = \mathbf{F}(w, x \uparrow_{\mathbf{B}(w,v,c)}^{\mathbf{A}(w,v,c)}).$$

We need to construct $\mathbf{F}$ satisfying (38) and (40) by carefully picking $\mathbf{A}$.

Let us see what condition we need to impose to $\mathbf{F}(w, x \uparrow_{\mathbf{B}(w,v,c)}^{\mathbf{A}(w,v,c)})$. We want it to be a candidate for $f(x, v) = \mathbf{F}(w, x \uparrow_{\mathbf{B}(w,v,c)}^{\mathbf{A}(w,v,c)})$, to be substituted in the following equation

$$\begin{aligned} t \mapsto f(X_t^\dagger, V_t) - \int_0^t \mathcal{A}^\dagger f(X_s^\dagger, V_s) ds \\ - \int_0^t \left. \frac{\partial}{\partial z} f(X_s^\dagger, z) \right|_{z=V_s} dV_s^c - \sum_{s \leq t} \left(f(X_s^\dagger, V_s) - f(X_s^\dagger, V_{s-}) \right) \end{aligned}$$

in order to get that the following process is a local martingale

$$t \mapsto \mathbf{F}(w, X_t^\dagger \uparrow_{\mathbf{B}(w,V_t,c)}^{\mathbf{A}(w,V_t,c)}).$$

So we need

$$(42) \quad \mathcal{A}^\dagger \mathbf{F}(w, x \uparrow_{\mathbf{B}(w,v,c)}^{\mathbf{A}(w,v,c)}) = 0 \quad \mathbf{B}_{ij}(w, v, c) < (x, i) < \mathbf{A}_{ij}(w, v, c), \forall j,$$

and we obtain this by choosing $\mathbf{F}$ as linear combination of transformed exit time matrices.

Then we need

$$(43) \quad \sum_{s \leq t} e^{-ws} \left(\mathbf{F}(w, X(s) \uparrow_{\mathbf{B}(w,\underline{X}(s),c)}^{\mathbf{A}(w,\underline{X}(s),c)}) - \mathbf{F}(w, X(s) \uparrow_{\mathbf{B}(w,\underline{X}(s^-),c)}^{\mathbf{A}(w,\underline{X}(s^-),c)}) \right) = 0.$$

6. MODULATED UPWARD ONE-SIDED PROBLEM

Let $\mathbf{a}$ be a vector in $\hat{\mathbb{R}}_+^n$, where $\hat{\mathbb{R}}_+ = \mathbb{R}_+ \cup \{\infty\}$. We define the set

$$\ell(\mathbf{a}) = \bigcup_{i=1}^n \{a_i\} \cup \{0, \infty\}$$

as the set of levels generated by the vector $\mathbf{a}$. In particular we denote by $|\ell(\mathbf{a})|$ the number of distinct levels generated by $\mathbf{a}$, and we denote the distinct orders levels by

$$(44) \quad 0 = \ell_0 < \ell_1 < \dots < \ell_{|\ell(\mathbf{a})|} = +\infty.$$

We also define the set $\ell(\mathbf{a}, \mathbf{b}) = \ell(\mathbf{a}) \cup \ell(\mathbf{b})$, with $\mathbf{b}$ another vector in $\hat{\mathbb{R}}_+^n$, and we denote in the same way the levels, as in (44), but now with the levels generated by both $\mathbf{a}$ and $\mathbf{b}$.

With $\mathcal{L} = \ell(\mathbf{a}, \mathbf{b})$, we have that the distinct levels partition the non-negative real line into $|\mathcal{L}|$ intervals,

$$(45) \quad \mathcal{I}_k = [\ell_k, \ell_{k+1}), \quad k = 0, \dots, |\mathcal{L}| - 1,$$

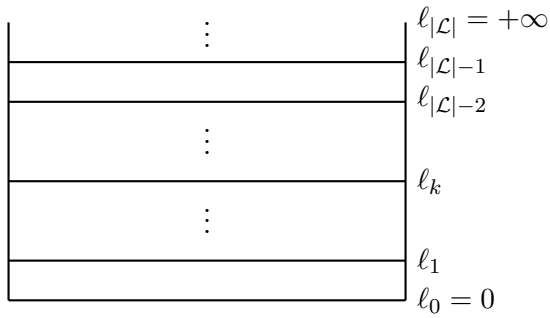
and we denote by E_k the set of indices i such that a_i is bigger than ℓ_k , while b_i is smaller or equal to ℓ_k , that is

$$(46) \quad E_k := \{i \in E : b_i \leq \ell_k < a_i\}.$$

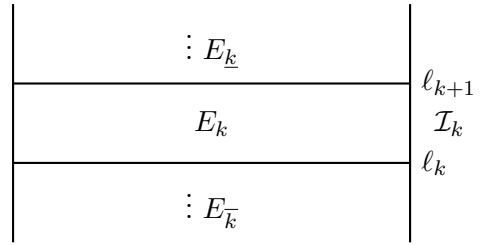
We also denote the sets

$$(47) \quad E_{\bar{k}} := \{i \in E : \ell_k < b_i < a_i\},$$

$$(48) \quad E_{\underline{k}} := \{i \in E : b_i < a_i \leq \ell_k\}.$$



(A) Ordered levels induced by $\mathbf{a}$ and $\mathbf{b}$.



(B) Active states on $\mathcal{I}_k$.

Now let $\mathcal{L} = \ell(\mathbf{a})$, and define the hitting time of the upward barrier $\mathbf{a}$ as

$$(49) \quad \tau_{\uparrow \mathbf{a}}(x, j) := \inf \{t > 0 : X(t) \geq a_{J(t)} \mid (X_0, J_0) = (x, j)\}.$$

Proposition 6.1. *For $\mathbf{a}' \leq \mathbf{a}$, the following equality holds in distribution:*

$$(50) \quad \tau_{\uparrow \mathbf{a}} = \tau_{\uparrow \mathbf{a}'} + \tau_{\uparrow \mathbf{a}}(X_{\tau_{\uparrow \mathbf{a}'}}, J_{\tau_{\uparrow \mathbf{a}'}}).$$

Proof. Since $\mathbf{a}' \leq \mathbf{a}$, we have that $a'_{J(t)} \leq a_{J(t)}$ for all $t \geq 0$, therefore,

$$(51) \quad \{t > 0 : X(t) \geq a'_{J(t)}\} \supseteq \{t > 0 : X(t) \geq a_{J(t)}\}.$$

By taking the infimum on both sides, we get that $\tau_{\uparrow \mathbf{a}'} \leq \tau_{\uparrow \mathbf{a}}$ a.s., and by the strong Markov property applied at $\tau_{\uparrow \mathbf{a}'}$, we have that (50) holds. $\square$

For any level ℓ_k , we define the truncated vector

$$(52) \quad \mathbf{a} \wedge \ell_k := (a_i \wedge \ell_k, i \in E),$$

and applying Proposition 6.1 with $\mathbf{a}' = \mathbf{a} \wedge \ell_k$, we get that

$$(53) \quad \tau_{\uparrow \mathbf{a}} = \tau_{\uparrow \mathbf{a} \wedge \ell_k} + \tau_{\uparrow \mathbf{a}}(X(\tau_{\uparrow \mathbf{a} \wedge \ell_k}), J(\tau_{\uparrow \mathbf{a} \wedge \ell_k})).$$

By applying the the Laplace-Stieltjes transform to the decomposition (53), we get the following proposition.

Proposition 6.2. *Let $x \leq \ell_k$, for some $k \in \{1, 2, \dots, |\ell(\mathbf{a})| + 1\}$, then*

$$(54) \quad \mathbf{Z}(x \uparrow^{\mathbf{a}}) = \mathbf{Z}(x \uparrow^{\mathbf{a} \wedge \ell_k}) \mathbf{Z}(\ell_k \uparrow^{\mathbf{a}}).$$

In addition, for any $x \geq 0$, $\mathbf{Z}(x \uparrow^{\mathbf{a}})$ is invertible.

Proof. By the definition of the Laplace-Stieltjes transform, we have that

$$\begin{aligned}
\mathbf{Z}(x \uparrow^{\mathbf{a}})_{ij} &= \mathbb{E}_{x,i} \left[e^{-w\tau \uparrow^{\mathbf{a}}} ; J(\tau \uparrow^{\mathbf{a}}) = j \right] \\
&= \mathbb{E}_{x,i} \left[\mathbb{E}_{x,i} \left[e^{-w\tau \uparrow^{\mathbf{a}}} ; J(\tau \uparrow^{\mathbf{a}}) = j \mid \mathcal{F}_{\tau \uparrow^{\mathbf{a}} \wedge \ell_k} \right] \right] \\
&= \mathbb{E}_{x,i} \left[e^{-w\tau \uparrow^{\mathbf{a}} \wedge \ell_k} \mathbb{E}_{X(\tau \uparrow^{\mathbf{a}} \wedge \ell_k), J(\tau \uparrow^{\mathbf{a}} \wedge \ell_k)} \left[e^{-w\tau \uparrow^{\mathbf{a}}} ; J(\tau \uparrow^{\mathbf{a}}) = j \right] \right] \\
&= \sum_h \mathbb{E}_{x,i} \left[e^{-w\tau \uparrow^{\mathbf{a}} \wedge \ell_k} ; J(\tau \uparrow^{\mathbf{a}} \wedge \ell_k) = h \right] \mathbb{E}_{a_h \wedge \ell_k, h} \left[e^{-w\tau \uparrow^{\mathbf{a}}} ; J(\tau \uparrow^{\mathbf{a}}) = j \right] \\
&= \sum_h \mathbf{Z}_{ih}(x \uparrow^{\mathbf{a} \wedge \ell_k}) \mathbf{Z}_{hj}(\ell_k \uparrow^{\mathbf{a}}),
\end{aligned}$$

where in the third equality we used the strong Markov property at $\tau \uparrow^{\mathbf{a}} \wedge \ell_k$, and in the last one we used the fact that $\mathbf{Z}_{hj}(a_h \wedge \ell_k \uparrow^{\mathbf{a}}) = \mathbf{Z}_{hj}(\ell_k \uparrow^{\mathbf{a}})$, for all $h \in E$. For the invertibility of $\mathbf{Z}(x \uparrow^{\mathbf{a}})$, we can use the standard local invertibility argument for continuous evolution families of matrices as in the proof of Theorem 2.1, see also Engel and Nagel [2000]. $\square$

We define the lowest level of the vector $\mathbf{a}$ as $\wedge_{\mathbf{a}} = \bigwedge(\mathbf{a}) := \min_{i \in E} a_i$. In particular, with $\mathcal{L} = \ell(\mathbf{a})$, $\ell_1 = \wedge_{\mathbf{a}}$. Then, the following result is immediate.

Proposition 6.3. *We have that, for any $x \in [0, \wedge_{\mathbf{a}})$,*

$$(55) \quad \mathbf{Z}(x \uparrow^{\mathbf{a} \wedge (\wedge_{\mathbf{a}})}) = \mathbf{Z}(x \uparrow^{\wedge_{\mathbf{a}}}),$$

where the matrix on the right-hand side has been computed in Section 2.

Next theorem shows how to compute then $\mathbf{Z}(x \uparrow^{\mathbf{a}})$ for $x \geq \wedge_{\mathbf{a}}$.

Theorem 6.4. *Assume that $\mathbf{Z}(x \uparrow^{\mathbf{a} \wedge \ell})$ is known for all $x < \ell$, for some $\ell > 0$, then there exists $p < 1$, not depending on ℓ , such that $\mathbf{Z}(x \uparrow^{\mathbf{a} \wedge (\ell/p)})$ can be computed for all $x < \ell/p$.*

In particular,

$$(56) \quad \mathbf{Z}(\ell \uparrow^{\mathbf{a} \wedge (\ell/p)}) = (\mathbf{I} - \mathbf{E}_k \mathbf{H}(\ell))^{-1} (\mathbf{I} + \mathbf{E}_k (\mathcal{E}(-(\ell/p - \ell)\mathbf{z}) - \mathbf{I})),$$

where the matrix $\mathbf{H}(x)$ is defined in (59), for $x \in [\ell, \ell/p)$.

Then, for $x \in (\ell, \ell/p)$

$$(57) \quad \mathbf{Z}(x \uparrow^{\mathbf{a} \wedge (\ell/p)}) = \mathbf{E}_k (\mathcal{E}(-(\ell/p - x)\mathbf{z}) + \mathbf{H}(x) \mathbf{Z}(\ell \uparrow^{\mathbf{a} \wedge (\ell/p)})) + (\mathbf{I} - \mathbf{E}_k).$$

and finally, for $x < \ell$,

$$(58) \quad \mathbf{Z}(x \uparrow^{\mathbf{a} \wedge (\ell/p)}) = \mathbf{Z}(x \uparrow^{\mathbf{a} \wedge \ell}) \mathbf{Z}(\ell \uparrow^{\mathbf{a} \wedge (\ell/p)}),$$

Proof. Without loss of generality, we may choose $p > 0$ such that $[\ell, \ell/p) \subset \mathcal{I}_k$ for some $k \in \{1, 2, \dots, |\ell(\mathbf{a})| + 1\}$ and in addition, such that $\max\{p_i : i \in E\} \leq p < 1$, so that for any $i \in E$, $p_i x < \ell$ for all $x \in [\ell, \ell/p)$.

By Proposition 6.2, we have that, for any $x < \ell$,

$$\mathbf{Z}(x \uparrow^{\mathbf{a} \wedge (\ell/p)}) = \mathbf{Z}(x \uparrow^{\mathbf{a} \wedge \ell}) \mathbf{Z}(\ell \uparrow^{\mathbf{a} \wedge (\ell/p)}),$$

so we are left with computing $\mathbf{Z}(x \uparrow^{\mathbf{a} \wedge (\ell/p)})$ only for $x \in [\ell, \ell/p)$.

For any $i \notin E_k$ and $x \in [\ell, \ell/p)$, $\mathbf{Z}_{ij}(x \uparrow^{\mathbf{a} \wedge (\ell/p)}) = \delta_{ij}$, so we need to compute $\mathbf{Z}_{ij}(x \uparrow^{\mathbf{a} \wedge (\ell/p)})$ only for $i \in E_k$ and $x \in [\ell, \ell/p)$.

Let $i \in E_k$ and $x \in [\ell, \ell/p)$. Then, let $t_i(x) = c_i(\ell/p - x)$, and $T \sim \text{Exp}(\lambda_i)$, then

$$\begin{aligned}
\mathbf{Z}_{ij}(x \uparrow^{\mathbf{a} \wedge (\ell/p)}) &= e^{-wt_i(x)} \delta_{ij} \mathbb{P}_{x,i}(T > t_i(x)) + \mathbb{E}_{x,i}[e^{-w\tau_{\uparrow \mathbf{a} \wedge (\ell/p)}}; J(\tau_{\uparrow \mathbf{a} \wedge (\ell/p)}) = j, T^i \leq t_i(x)] \\
&= e^{-(w+\lambda_i)t_i(x)} \delta_{ij} + \sum_h q_{ih} \int_0^{t_i(x)} \mathbf{Z}_{hj}((x+t/c_i)p_i \uparrow^{\mathbf{a} \wedge (\ell/p)}) \lambda_i e^{-(w+\lambda_i)t} dt \\
&= e^{-z_i(\ell/p-x)} \delta_{ij} + \sum_h q_{ih} \gamma_i e^{\frac{z_i}{p_i}x} \int_{p_i x}^{p_i \ell/p} \mathbf{Z}_{hj}(y \uparrow^{\mathbf{a} \wedge (\ell/p)}) e^{-\frac{z_i}{p_i}y} dy \\
&= e^{-z_i(\ell/p-x)} \delta_{ij} + \sum_{h'} \gamma_i e^{\frac{z_i}{p_i}x} \int_{p_i x}^{p_i \ell/p} \sum_h q_{ih} \mathbf{Z}_{hh'}(y \uparrow^{\mathbf{a} \wedge (\ell/p)}) e^{-\frac{z_i}{p_i}y} dy \mathbf{Z}_{h'j}(\ell \uparrow^{\mathbf{a} \wedge (\ell/p)}) \\
&= e^{-z_i(\ell/p-x)} \delta_{ij} + \sum_{h'} \mathbf{H}_{ih'}(x) \mathbf{Z}_{h'j}(\ell \uparrow^{\mathbf{a} \wedge (\ell/p)})
\end{aligned}$$

where, for any $i, j \in E$ and $x \in [\ell, \ell/p)$, the quantity

$$(59) \quad \mathbf{H}_{ij}(x) := \gamma_i e^{\frac{z_i}{p_i}x} \int_{p_i x}^{p_i \ell/p} \sum_h q_{ih} \mathbf{Z}_{hj}(y \uparrow^{\mathbf{a} \wedge (\ell/p)}) e^{-\frac{z_i}{p_i}y} dy,$$

can be computed since $\mathbf{Z}(y \uparrow^{\mathbf{a} \wedge (\ell/p)})$ is known for all $y < \ell$ and $p_i x < p_i \ell/p \leq \ell$.

Let $\mathbf{E}_k = ((i \in E_k) \delta_{ij})_{i,j}$ be the diagonal matrix with entries δ_{ij} if $i \in E_k$ and 0 otherwise, then we can write the previous equation in matrix form as

$$\mathbf{Z}(x \uparrow^{\mathbf{a} \wedge (\ell/p)}) = \mathbf{E}_k (\mathcal{E}(-(\ell/p - x)\mathbf{z}) + \mathbf{H}(x) \mathbf{Z}(\ell \uparrow^{\mathbf{a} \wedge (\ell/p)})) + (\mathbf{I} - \mathbf{E}_k).$$

and in particular, for $x = \ell$,

$$\mathbf{Z}(\ell \uparrow^{\mathbf{a} \wedge (\ell/p)}) = (\mathbf{I} - \mathbf{E}_k \mathbf{H}(\ell))^{-1} (\mathbf{I} + \mathbf{E}_k (\mathcal{E}(-(\ell/p - \ell)\mathbf{z}) - \mathbf{I})).$$

□

7. MODULATED UPWARD TWO-SIDED PROBLEM

Proposition 7.1. *Let $\mathbf{a} \leq b/p$ then for any $x \in [b, \wedge_{b/p})$, we have*

$$(60) \quad \mathbf{L}(x \uparrow_b^{\mathbf{a}}) = \mathcal{E}(-(\mathbf{a} - x)\mathbf{z}),$$

in particular, for $x = b$, $\mathbf{L}(b \uparrow_b^{\mathbf{a}}) = \mathcal{E}(-(\mathbf{a} - b)\mathbf{z})$.

Proof. Assume that $(x_0, J_0) = (x, i)$, then, since $a_i p_i \leq b$, we have that, with $T \sim \text{Exp}(\lambda_i)$, $\{\tau_{\uparrow \mathbf{a}} < \tau_{\downarrow b}\} = \{c_i T > (a_i - x)\}$. Therefore

$$(61) \quad \mathbf{L}_{ij}(x \uparrow_b^{\mathbf{a}}) = \delta_{ij} \mathbb{P}_{x,i}(\tau_{\uparrow \mathbf{a}} < \tau_{\downarrow b}) = \delta_{ij} \mathbb{P}_{x,i}(c_i T \geq (a_i - x)) = \delta_{ij} \exp\{-(z_i/p)(a_i - x)\},$$

and the result follows. □

Proposition 7.2. *Let $b \leq x < \mathbf{a}$, and let $x' \geq x$, then*

$$(62) \quad \mathbf{L}(x \uparrow_b^{\mathbf{a}}) = \mathbf{L}(x \uparrow_b^{\mathbf{a} \wedge x'}) \mathbf{L}(x' \uparrow_b^{\mathbf{a}}).$$

In addition, for any $x \geq b$, $\mathbf{L}(x \uparrow_b^{\mathbf{a}})$ is invertible.

Proof. Since $\tau_{\uparrow \mathbf{a}} \geq \tau_{\uparrow \mathbf{a} \wedge x'}$ on $\{\tau_{\uparrow \mathbf{a}} \leq \tau_{\downarrow b}\}$ by applying the strong Markov property at time $\tau_{\uparrow \mathbf{a} \wedge x'}$ on the set $\{\tau_{\uparrow \mathbf{a}} \leq \tau_{\downarrow b}\}$, and using the fact that the process starts anew in $(a_{J(\tau_{\uparrow x})} \wedge x', J(\tau_{\uparrow x}))$, we get the result. For the invertibility, we use the standard local invertibility argument for continuous evolution families of matrices as in the proof of Theorem 2.1, see also Engel and Nagel [2000]. □

REFERENCES

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